

## MCGUIRE AFB, NJ, USA

WMO: 724096

Lat: 40.017N

Lon: 74.600W

Elev: 131

StdP: 14.63

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14706

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 10.5       | 15.1 | -3.7                          | 4.5 | 14.0 | 0.9 | 5.8 | 18.2 | 27.6                  | 35.0 | 25.2 | 32.6 | 9.3                   | 280  | 0.492 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 19.5                   | 92.9            | 75.7 | 90.3 | 74.6 | 87.8 | 73.4 | 78.7                | 87.5 | 77.3 | 86.1 | 75.9 | 83.8 | 10.5                 | 250  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 76.6                            | 139.5 | 82.6 | 74.8 | 131.2 | 81.2 | 73.3 | 124.2 | 79.6 | 42.3          | 87.6 | 40.7 | 85.9 | 39.3 | 83.7 | 86.4              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |      |            |      |            |      |       |     |
|-------------------|------|------|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|------|------------|------|------------|------|-------|-----|
|                   |      |      | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |      | n=20 years |      | n=50 years |      |       |     |
| 1%                | 2.5% | 5%   | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max  | Min        | Max  | Min        | Max  |       |     |
| (a)               | (b)  | (c)  | (e)                        | (f) | (g)                | (h) | (i)                                                | (j) | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |       |     |
| (4) 23.5          | 20.1 | 18.0 | DB                         | 4.9 | 97.5               | 4.7 | 3.6                                                | 1.5 | 100.1      | -1.2 | 102.2      | -3.8 | 104.2      | -7.2 | 106.8 | (4) |
| (5)               |      |      | WB                         | 3.7 | 81.5               | 4.5 | 2.2                                                | 0.5 | 83.1       | -2.2 | 84.4       | -4.7 | 85.6       | -8.0 | 87.2  | (5) |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 54.7   | 32.1  | 34.9  | 42.1  | 52.9  | 62.3  | 71.3  | 76.5  | 74.6  | 67.9  | 56.8  | 46.1  | 37.7 |
|                                                                                  | DBStd       | 17.28  | 10.61 | 9.55  | 9.12  | 8.62  | 7.68  | 6.56  | 5.12  | 5.17  | 6.51  | 8.02  | 8.30  | 9.12 |
|                                                                                  | HDD50       | 1926   | 563   | 431   | 279   | 60    | 2     | 0     | 0     | 0     | 0     | 22    | 172   | 397  |
|                                                                                  | HDD65       | 4860   | 1020  | 844   | 712   | 378   | 147   | 20    | 0     | 2     | 43    | 279   | 569   | 846  |
|                                                                                  | CDD50       | 3642   | 8     | 8     | 33    | 147   | 384   | 638   | 822   | 762   | 537   | 233   | 54    | 16   |
|                                                                                  | CDD65       | 1100   | 0     | 0     | 1     | 15    | 64    | 208   | 357   | 299   | 130   | 25    | 1     | 0    |
|                                                                                  | CDH74       | 9807   | 0     | 1     | 18    | 187   | 662   | 1875  | 3370  | 2545  | 968   | 174   | 7     | 0    |
|                                                                                  | CDH80       | 3650   | 0     | 0     | 3     | 57    | 211   | 683   | 1428  | 941   | 287   | 38    | 2     | 0    |
| Wind                                                                             | WSAvg       | 8.1    | 9.4   | 9.5   | 10.0  | 9.4   | 7.8   | 7.2   | 6.7   | 6.2   | 6.7   | 7.4   | 8.1   | 8.6  |
| Precipitation                                                                    | PrecAvg     | 45.1   | 3.5   | 2.5   | 4.0   | 3.5   | 3.5   | 4.1   | 4.4   | 4.8   | 4.0   | 3.7   | 3.0   | 4.0  |
|                                                                                  | PrecMax     | 57.5   | 7.0   | 4.6   | 8.5   | 10.0  | 6.8   | 8.6   | 10.0  | 14.6  | 8.3   | 10.8  | 6.5   | 8.0  |
|                                                                                  | PrecMin     | 31.7   | 1.7   | 0.6   | 1.2   | 1.3   | 0.9   | 1.3   | 1.1   | 1.1   | 0.8   | 0.7   | 0.8   | 1.1  |
|                                                                                  | PrecStd     | 6.1    | 1.3   | 1.0   | 1.8   | 1.8   | 1.3   | 2.1   | 1.7   | 2.7   | 2.1   | 2.1   | 1.4   | 1.6  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 65.6  | 68.1  | 75.6  | 87.0  | 90.3  | 94.3  | 97.0  | 95.5  | 91.3  | 84.3  | 72.9  | 67.8 |
|                                                                                  |             | MCWB   | 58.9  | 57.9  | 60.9  | 66.4  | 71.2  | 76.4  | 76.3  | 76.4  | 74.4  | 69.5  | 62.4  | 62.2 |
|                                                                                  | 2%          | DB     | 59.9  | 61.1  | 68.3  | 79.3  | 86.0  | 90.5  | 93.4  | 91.0  | 87.1  | 79.0  | 68.1  | 61.8 |
|                                                                                  |             | MCWB   | 56.5  | 53.5  | 56.2  | 62.7  | 70.2  | 74.7  | 76.0  | 75.5  | 72.2  | 67.5  | 60.3  | 58.0 |
|                                                                                  | 5%          | DB     | 53.9  | 55.0  | 62.7  | 73.3  | 81.5  | 87.5  | 90.7  | 88.1  | 83.1  | 74.1  | 64.0  | 57.2 |
|                                                                                  |             | MCWB   | 49.9  | 48.7  | 53.4  | 60.0  | 68.0  | 73.4  | 75.2  | 74.1  | 71.0  | 65.1  | 58.2  | 53.5 |
|                                                                                  | 10%         | DB     | 47.8  | 49.8  | 57.4  | 68.2  | 77.1  | 84.0  | 87.8  | 85.6  | 79.7  | 70.3  | 60.3  | 52.5 |
|                                                                                  |             | MCWB   | 43.7  | 43.9  | 50.2  | 57.4  | 65.3  | 71.0  | 74.1  | 72.9  | 68.9  | 63.2  | 54.7  | 48.4 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 61.6  | 61.6  | 63.4  | 69.1  | 74.5  | 79.1  | 81.4  | 80.7  | 77.7  | 73.5  | 66.5  | 64.0 |
|                                                                                  |             | MCDB   | 63.8  | 66.3  | 71.6  | 81.2  | 85.3  | 88.9  | 89.5  | 88.5  | 86.3  | 78.6  | 69.3  | 65.2 |
|                                                                                  | 2%          | WB     | 56.8  | 55.5  | 59.2  | 66.0  | 72.2  | 77.0  | 79.1  | 78.5  | 75.5  | 70.4  | 63.4  | 59.1 |
|                                                                                  |             | MCDB   | 60.0  | 58.8  | 66.2  | 75.2  | 82.3  | 86.9  | 88.2  | 86.4  | 81.6  | 75.7  | 66.0  | 61.1 |
|                                                                                  | 5%          | WB     | 50.4  | 49.4  | 55.1  | 63.0  | 70.0  | 75.2  | 77.5  | 76.8  | 73.6  | 67.8  | 59.5  | 54.5 |
|                                                                                  |             | MCDB   | 52.9  | 53.7  | 60.7  | 70.5  | 78.3  | 84.1  | 86.7  | 84.4  | 79.2  | 72.4  | 62.8  | 57.3 |
|                                                                                  | 10%         | WB     | 44.4  | 44.1  | 50.6  | 59.5  | 67.7  | 73.3  | 76.1  | 75.1  | 72.0  | 64.6  | 55.8  | 49.4 |
|                                                                                  |             | MCDB   | 46.9  | 48.8  | 56.8  | 66.5  | 74.1  | 80.9  | 84.6  | 81.8  | 77.3  | 68.8  | 59.5  | 51.9 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 15.8  | 17.5  | 19.0  | 21.1  | 20.3  | 19.8  | 19.5  | 19.1  | 19.6  | 19.9  | 18.8  | 16.0 |
|                                                                                  |             | MCDBR  | 22.4  | 25.9  | 27.5  | 28.3  | 26.2  | 23.2  | 22.5  | 21.6  | 23.1  | 23.8  | 23.4  | 20.9 |
|                                                                                  |             | MCWBR  | 18.8  | 19.3  | 17.9  | 14.8  | 12.8  | 10.5  | 8.5   | 9.0   | 10.8  | 13.3  | 16.4  | 18.7 |
|                                                                                  | 5% WB       | MCDBR  | 21.1  | 23.5  | 24.6  | 24.0  | 23.0  | 21.3  | 19.9  | 18.8  | 18.5  | 19.7  | 18.7  | 20.0 |
|                                                                                  |             | MCWBR  | 19.4  | 20.2  | 18.6  | 14.9  | 12.4  | 10.9  | 9.1   | 8.9   | 10.3  | 13.3  | 16.0  | 19.2 |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.312  | 0.320 | 0.357 | 0.386 | 0.432 | 0.493 | 0.493 | 0.486 | 0.404 | 0.360 | 0.329 | 0.310 |      |
|                                                                                  | taud        | 2.473  | 2.446 | 2.397 | 2.340 | 2.243 | 2.111 | 2.125 | 2.135 | 2.337 | 2.468 | 2.512 | 2.527 |      |
|                                                                                  | Ebn at Noon | 271    | 285   | 285   | 282   | 270   | 252   | 251   | 249   | 265   | 267   | 263   | 264   |      |
|                                                                                  | Edh at Noon | 25     | 29    | 34    | 39    | 43    | 50    | 49    | 47    | 36    | 29    | 24    | 22    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 615    | 905   | 1201  | 1548  | 1714  | 1881  | 1920  | 1660  | 1381  | 981   | 698   | 536   |      |
|                                                                                  | RadStd      | 54     | 97    | 98    | 134   | 197   | 160   | 92    | 109   | 139   | 99    | 62    | 52    |      |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (66 neighbors) | +0.72 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | +214  | +122  |

Nomenclature: See separate page